

MOHAMED ATEF

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Experience

Artificial Intelligence Engineer Intern – EGRONX

Aug 2025 – Oct 2025 | Remote

- Built Retrieval-Augmented Generation (RAG) applications using LangChain, FastAPI, and Docker within an Agile Scrum team, serving as Scrum Master to coordinate sprint planning and daily stand-ups.
- Built a RAG Q&A chatbot grounded in 7 organizational bylaws, exposed via high-performance FastAPI microservices, delivering accurate, citation-backed responses achieving 96.5% accuracy on a benchmark of 300 curated questions on Hugging Face Spaces.

Computer Vision Intern – Celula

Aug 2025 – Sep 2025 | Remote

- Developed YOLO-based anti-theft object detection and water segmentation pipelines for 12-band multispectral satellite imagery, enabling real-time inference through a Flask web application.

Native Android Application Trainee – Banque Misr

Aug 2024 – Sep 2024 | Egypt

- Developed native Android applications using Kotlin and Jetpack Compose following MVVM architecture pattern.
- Created the Spedo Transfer Application, ranked among the top 5 projects across the entire internship cohort.

Education

Cairo University – Cairo, Egypt

Jun 2026

B.Sc. in Computer Science and Artificial Intelligence

GPA: 3.53/4.00 (Excellent)

Ranked among the Top 20 graduates out of 180 students in the AI Department

Competitions & Activities

AI Developer – CU-AI Nexus Hackathon

2025 | Cairo University

- Won 2nd place at Cairo University and 4th place overall across participating universities.
- Built AI-powered mentor assessment using semantic search over learning resources; matched mentees via similarity.
- Developed an AI interview preparation assistant featuring technical Q&A generation, body posture analysis using MediaPipe, Whisper for filler-word detection and spoken interview feedback.

Technical Skills

Programming: Python, SQL, JavaScript, TypeScript, Kotlin, C++

AI/ML: PyTorch, Transformers, LangChain, LangGraph, TensorFlow, FAISS, MediaPipe, Whisper, BM25

Deployment: FastAPI, Docker, Git, GitHub Actions, Vercel, Hugging Face Spaces

Domains: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language Processing, Agentic AI, Computer Vision, Deep Learning

Languages: Arabic (Native), English (Fluent)

Projects

🐙 Arabic Grammarly System (BAYAN) — Highest-Rated Graduation Project in Committee (A+) — NLP, PyTorch

- Built an Arabic AI writing assistant providing grammar correction, spelling correction, punctuation restoration, text summarization, Dialect-to-MSA conversion, Quran text verification, and auto-complete through a web application and Chrome extension.
- Fine-tuned Transformer encoder-decoder models using Arabic NLP datasets (including LANS), achieving a ROUGE-1 score of 33.94 on summarization evaluation.
- Designed an evaluation benchmark of 360 Arabic sentences, achieving 94.5% accuracy, and authored unit tests covering core NLP modules to ensure prediction reliability.
- Containerized the application with Docker and deployed on Hugging Face Spaces for cloud inference.

🐙 RAG Chatbot for Islamic Content | RAG, FAISS, Validation Pipeline

- Built a RAG system answering questions on Quran, Tafsir, Hadith, and Prophet stories, with monitoring in place to track model latency, system health, and retrieval relevance after deployment on Vercel.
- Engineered a 7-layer validation pipeline ensuring factual accuracy and eliminating hallucination in sensitive religious content.

🐙 Multi-Agent Document Q&A System | LangChain, LangGraph, Agentic AI

- Designed a collaborative 3-agent workflow using LangGraph to process documents and generate citation-backed answers through coordinated agent execution.
- Implemented a GitHub Actions CI/CD pipeline automating unit tests, evaluation, Docker image builds, and deployment.